

Project 1: FSM Front-End Design

Requirements:

Practice the topic of front-end design for an FSM: (Work as teams of **3 students**).

1. Design the FSM diagram of a **hh:mm 4-digit 24h-format** watch **controller**.
It has **4 modes**:
normal: displays time,
set time: set 1 digit at a time from left to right,
set alarm: set 1 digit at a time from left to right,
stop-watch: perform start-stop and split time, measured as mm:ss.
It has **only 2 push-buttons**: **mode**: press to go sequentially through the 4 modes and through the digits. **set**: press to increment the digits. Use the CASIO manual below as a guide but only implement what is required here. Assume all needed peripherals and their interfaces.
2. Write Verilog code to implement the design.
3. Develop test strategy (provide test scenario).
4. Design one testbench that automatically check the correctness of outputs.
5. Simulate the testbench and comment on results.

Deliverables:

A **single** zip file **<Gxx_project1>.zip** containing a top-level directory named <Gxx> under which **only TWO** components:

- I. A **single** pdf file named <Gxx_project1>.pdf containing the following:
 1. Cover page. Contains: project title + students name + emails + department
 2. Chapter 1: Introduction. Contains: project idea + problem description.
 3. Chapter 2: FSM design. Contains: design details, system diagram and FSM diagram.
 4. Chapter 3: RTL design. Contains: RTL codes and comments.
 5. Chapter 4: TB design. Contains: test scenario, TB code, and comments.
 6. Chapter 5: Simulation. Contains: simulation waveforms and comments.
 - II. A **directory** named <files> containing all source and output files.
- xx is your 2-digit team group number as identified by the TA.

Submission:

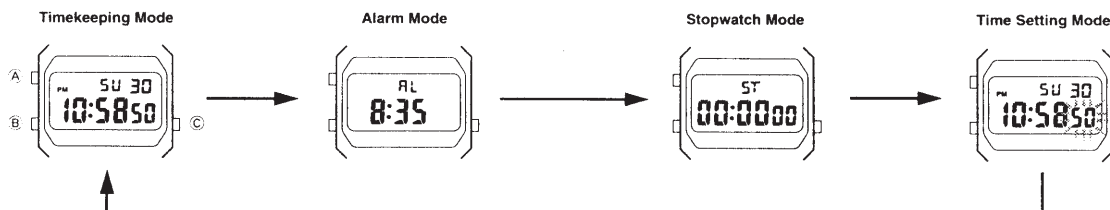
Submit to course LMS site before end of day 19th of December 2025.

All members of the group must submit the same file on LMS.

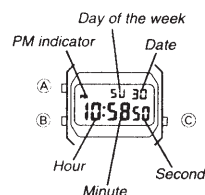
OPERATION CHART:MODULE QW-1275

GENERAL GUIDE

- Press **B** to change from mode to mode.
- After you perform an operation in any mode, pressing **B** returns to the Timekeeping Mode.



TIMEKEEPING MODE

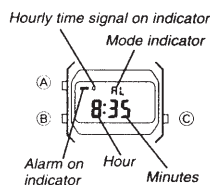


- In the Timekeeping Mode, press **C** to switch between the 12-hour and 24-hour formats.
- Day of the week is displayed as follows:
 SU: Sunday MO: Monday TU: Tuesday
 WE: Wednesday TH: Thursday FR: Friday
 SA: Saturday

About the backlight

- Holding down **A** at any time illuminates the display.
- The backlight of this watch employs an electro-luminescent (EL) light, which loses its illuminating power after very long term use.
- Frequent use of the backlight shortens the battery life.

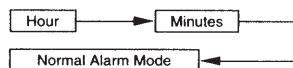
ALARM MODE



When the Daily Alarm is switched on, the alarm sounds for 20 seconds at the preset time each day. When the Hourly Time Signal is switched on, the watch beeps every hour on the hour.

To set the alarm time

1. Hold down **A** while in the Alarm Mode until the hour digits start to flash on the display. The hour digits flash because they are *selected*.
2. Press **A** to change the selection in the following sequence.



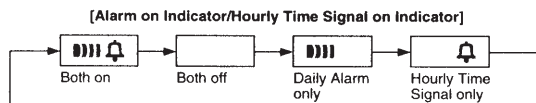
- Once you reach the normal Alarm Mode, you have to hold down **A** again until the hour digits start to flash.
- 3. Press **C** to increase the selected digits. Holding down **C** changes the number at high speed.
- The format (12-hour and 24-hour) of the alarm time matches the format you select for normal timekeeping.
- When setting the alarm time using the 12-hour format, take care to set the time correctly as morning or afternoon (PM).
- 4. After you set the alarm time, press **A** to return to the Alarm Mode. At this time the Daily Alarm is switched on automatically.

To stop the alarm

Press **A** to stop the alarm after it starts to sound.

To switch the Daily Alarm and Hourly Time Signal on and off

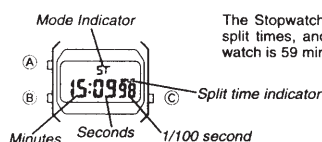
Press **C** while in the Alarm Mode to change the status of the Daily Alarm and Hourly Time Signal in the following sequence.



To test the alarm

Hold down **C** while in the Alarm Mode to sound the alarm.

STOPWATCH MODE

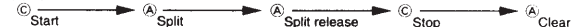


The Stopwatch Mode lets you measure elapsed time, split times, and two finishes. The range of the stopwatch is 59 minutes, 59.99 seconds.

(a) Elapsed time measurement



(b) Split time measurement



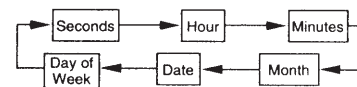
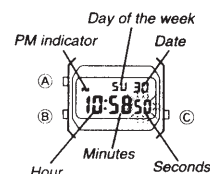
(c) Split time and 1st-2nd place times



TIME SETTING MODE

To set the time and date

1. Use **B** to enter the Time Setting Mode.
2. Press **A** to change the selection in the following sequence.



3. While the seconds digits are selected (flashing), press **C** to reset the seconds to "00". If you press **C** while the seconds count is in the range of 30 to 59, the seconds are reset to "00" and 1 is added to the minutes. If the seconds count is in the range of 00 to 29, the minutes count is unchanged.
4. While any other digits (besides seconds) are selected (flashing), press **C** to increase the number. While the day of the week is selected, pressing **C** advances to the next day. Holding down **C** changes the current selection at high speed.
5. After you set the time and date, press **B** to return to the Timekeeping Mode.

- If you do not operate any button for a few minutes while a selection is flashing, the flashing stops and the watch goes back to the Timekeeping Mode automatically.